"How do I make games?" A Path to Game Development

by Geoff Howland

Starcraft, Everquest, Quake and Counter-Strike were all made by teams of professionals who had budgets usually million dollars plus. All of these games were made by people with a lot of experience at making games. They did not just decide to make games which turned out mega-hits, they started out small and worked their way up.

So where do I start? Tetris is the perfect game to begin your journey on the path to becoming an able-bodied game developer. Why? Because Tetris has all the individual components that ALL games share in common. It has a game loop (the process of repeating over and over until the game is quit). The game loop reads in input, processes the input, updates the elements of the game (the falling tetraminos), and checks for victory/loss conditions. Also, you don't have to be an artist to make a good-looking Tetris game. Anyone who can draw a block, which is everyone with a paint program, can make a commercial quality version of Tetris. Something I need to mention is that when you make your Tetris game, you can't call it "Tetris". However, this means nothing to you if you call your game "The Sky is Falling", or anything without a "tris" in it.

What's next? After you have totally, completely, absolutely finished your version Tetris, you are ready for your next challenge: Breakout. It is a similar game, but adds in much more advanced collision detection than was necessary in Tetris. Here you will need to add some simple deflection physics of the ball rebounding off different portions of the paddle and the blocks. Level layout also becomes an issue in Breakout, and in order to have more than one level you will need to come up with a way to save the maps. This deals with another component found in all larger games, which is saving and loading resources and switching levels.

After you finish your Breakout masterpiece you should move on to making Pac-Man. Pac-Man is an evolutionary step because it adds in the element of enemy artificial intelligence (AI). You may not have been aware of this, but in the original Pac-Man the four different ghosts had different goals to trying to defeat you as a team. The aggressor would try to follow the shortest path to you, making you directly avoid him. The interceptor would try to go to a junction that was closest to where you would have to move to avoid the aggressor. A second interceptor would try to stay more towards the middle and try to cut you off from using the tunnel through the sides. The last ghost would sort of wander aimlessly about which often kept him staying in a section you needed to finish the map.

Pac-Man also increases the complexity of maps, and adds a good deal more flexibility for using sounds, as sound was certainly a crucial element to the success of Pac-Man. (After all, what would Pac-Man be without some sort of "wakka-wakka" sound?)

The last game I suggest you should create is a side scroller, such as Super Mario Brothers, where you can jump on multiple platforms, shoot, duck and interact with enemies. Now you must make a screen that is capable of scrolling in at least two directions, if not four, and deal with screen clipping, which can have a bit of a learning curve. You must also work on the physics of any jumping, bouncing of the character or shooting projectiles.

There will additionally need to be a lot more enemies than before, and you will need to keep track of their current game state (alive/dead, active/inactive), by whether they are on the screen or have already been dealt with. The level editor should be capable of placing tiles, scrolling through tiles, scrolling over the map, choosing tiles as brushes, cycling through the brushes, cutting and pasting, an undo, and placing enemies. I would also suggest making backups of previously saved maps, as it is often easier to just back things up by versions, than redrawing them.

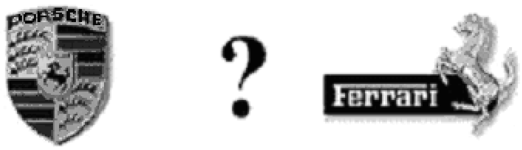
Finally, the side scroller has a real victory condition! When you get to the end of the side scroller, you have actually GONE somewhere, so you can add on a story to progress through the game as well (and don't forget some sort of fireworks on the screen for the end of a level, so that the player has a sense of accomplishment and a REAL show of fireworks for beating the game. Merely putting the words "You Have Won!" on the screen when a player has spent endless hours trying to beat your game is anti-climactic).

But, these games are stupid! Actually, these games clearly show the basis for ALL games' gameplay. Throw a fancy 3D interface over a shooter and it's still a shooter. You could create the same game in a 2D overhead view and the gameplay would be coded exactly the same.

Is it stupid to be able to make a game with EXACTLY the same controls, responses and enemies as Quake? If you remove the 3D interface, and look at what is really happening from a directly overhead view, does it still seem as out of reach?

One thing that you need to clarify to yourself before starting anything, is what you want out of it. Do you want to make games, or just duplicate the technology in Quake? If all you are interested in is the technology, then skip all the games stuff and get started on graphics technology.

I made my game, now where's my Ferrari? Sorry, one game, two games, five games probably won't cut it. Last year there were 3,500 games released on the PC, and only a few handfuls made back a large portion of cash. Most of those that did weren't made by small groups who were self-funded, they were funded by large publishers and probably had multi-million dollar budgets, and definitely near or well over million dollar advertising campaigns. This isn't a world you can't join though, it just takes a good deal of time and experience and track record of making quality games, that hopefully sell well, to give publishers confidence in your team, so that they will entrust you with this kind of financial responsibility.



Decisions, decisions. However, there is more to making a living of games than the multi-million dollar budgets. Just have an understanding of what you really want out of making games and then concentrate on making that come true.